Remote Logging using Rsyslog and MySQL

Variously described by its creators as the Swiss army knife of logging or the rocket-fast system for processing logs, Rsyslog offers great security features, high performance and a modular design. It can receive log inputs from various sources, transform them and output them to the desired locations, from which they can be processed.

The volume of event logs generated per system is increasing daily, mostly due to sophistication in hardware and software, which requires greater monitoring for cases of abnormality and second, as a security measure that demands more event logs to understand external intervention in the system. Centralising this voluminous log data to a dedicated log server over the network is essential even for a medium scale organisation, in order to have an overall view of what is happening across the IT infrastructure. This article will help readers who are new to this subject, to configure a small and centralised remote-logging set-up.

Rsyslog is an open source software framework for log processing on GNU/ Linux and UNIX platforms. It is a multi-threaded, network-aware, modular, highly customisable system that is suitable for any currently conceivable logging scenario.

Let us consider some of the typical logging scenarios of Rsyslog.

1. Single host configuration

This is the simplest configuration in which the Rsyslog server stores the log messages in the same machine, either in a disk file or database see Figure 1.

2. Dual host configuration

Here, two Rsyslog servers are physically separated over an IP network via WAN or LAN and the second Rsyslog server acts as a dedicated log server see Figure 2.

3. Dual host with a remote database server

The database server is also physically separated from the log server over an IP network see Figure 3.

4. Triple host configuration with a Rsyslog relay server

The middle Rsyslog server relays the log messages coming from Host A to Host C, the log server. Other possible configurations are more or less some combination of these four basic forms see Figure 4.

Now let's configure our Rsyslog server in the dual host configuration discussed above. I have used the Ubuntu 14.04 LTS version for this set-up, though you can select any distribution of your choice. We have two hosts—Ubuntu1404lts1 (IP:10.0.6.100) and Ubuntu1404lts2 (IP:10.0.6.101), where the former forwards all its log messages to the latter via TCP/IP, which in turn stores all received log messages to a local MySQL database. Let's install the required packages for our set-up. Rsyslog version 6 and above and any MySQL version will suffice. If any of these packages are already installed in your system, you can skip that step. To check, you can run the following command at the shell:

```
dpkg-query -s <package-name>
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